

# MONISHKA DAMODAR NAIDU

GenAI Engineer | Backend AI Engineer

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## PROFESSIONAL SUMMARY

Results-driven GenAI & Backend AI Engineer with hands-on experience building and deploying production-grade AI systems on Google Cloud Platform. Proven track record in RAG pipelines, computer vision, multilingual OCR, and LLM integration. backend engineer and client-facing technical lead in current engagement, responsible for GCP service deployments, Cloud infrastructure, and end-to-end CI/CD delivery. Experienced in translating complex AI capabilities into real business for enterprise and government

## TECHNICAL SKILLS

### Languages:

**GenAI & LLMs:** Generative AI, RAG Systems, Chunking Strategies, Gemini LLM, Mistral 7B, LoRA Fine-tuning, LLM Prompt Engineering, LangChain, LangGraph,

**GCP AI Services:** Google Document AI, Vertex AI, Cloud Vision API, Translation API, Speech-to-Text API, Text-to-Speech

**Computer Vision:** OpenCV, Image Optimization, Face Recognition, OCR, Signature

**Cloud & Infrastructure:** Google Cloud Platform (GCP), Cloud Run, Cloud Functions, GCP Data Store, Terraform, Terraform Docker (basic), CI/CD

**Frameworks & APIs:** FastAPI,

**DevOps & Version Control:** Git, GitHub, CI/CD, Terraform

**Soft Skills:** Client-facing technical communication, stakeholder presentations, cross-functional collaboration, production

## EXPERIENCE

### Tech Mahindra | GenAI & Backend AI Engineer

Nagpur, India • IT Services & Consulting Jul 2023 –

#### Client: American Client | Jul 2023 – Sep 2023 (POC) & Apr 2025 –

##### American Client

- Sole backend engineer responsible for deploying and managing all AI services on Google Cloud Platform (GCP), including Run, Cloud Functions, and GCP Data
- Architected and deployed infrastructure using Terraform and Terraform Cloud, managing full production environment zero-downtime
- Serve as the primary client-facing technical point of contact on all backend architecture, deployment decisions, and AI — directly communicating with US-based
- Built a production RAG system using Gemini LLM and GCP Data Store, reducing information retrieval time for managers workers by 40%, eliminating manual lookups across fragmented knowledge
- Engineered multilingual OCR pipelines using Google Document AI across 30+ languages, saving 100+ manual hours per data entry across global
- Integrated and automated Google APIs for Translation, Speech-to-Text, and Text-to-Speech, enabling multilingual AI for a global
- Automated validation testing for 40 target languages, achieving 95–99% accuracy before production deployment, reliability at
- Integrated AI services with the frontend application and managed Git workflows, ensuring seamless full-stack
- Collaborated with cross-functional teams across engineering, product, and operations to assess scalability and real-adoption of GenAI

*Tech Stack: Python, Gemini LLM, Google Document AI, GCP (Cloud Run, Cloud Functions, GCP Data Store), Terraform, Cloud, Translation API, Speech-to-Text API, Text-to-Speech API, RAG, CI/CD,*

#### Client: Dubai Government Project | Nov 2024 – Apr 2025

##### Dubai Government Engagement

- Delivered ML solutions for a UAE government client, designing an image optimization pipeline to validate image processing

- Designed and implemented production image optimization pipelines using OpenCV, improving input quality for models in high-stakes government
- Applied computer vision techniques for face recognition and image processing in live production environments with compliance
- Presented technical analysis directly to the client, leading to a data-driven decision to descope the signature verification feature resulting in a 15% reduction in project scope and significant budget
- Deployed AI services via Flask APIs and integrated with CI/CD pipelines to ensure scalability, reliability, and production

*Tech Stack: Python, OpenCV, Flask, Face Recognition, Image Processing, ML Models, CI/CD*

## PROJECTS

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### JobCrew AI · Personal Project (In Progress)

- Architected a multi-agent AI system using CrewAI with 5 specialized agents (Resume Analyzer, Job Researcher, Cover Writer, Interview Coach, Application Tracker) coordinated by a central orchestrator
- Implemented vector-based memory and retrieval using ChromaDB, enabling agents to reference past job applications, context, and role-specific insights across
- Integrated Groq-hosted Llama 3 as the inference backend, achieving low-latency agent responses suitable for job-search
- Designed inter-agent communication and task delegation patterns using LangGraph-style orchestration logic within demonstrating production-grade multi-agent system

*Tech Stack: Python, CrewAI, LangGraph, ChromaDB, Groq, Llama 3, Multi-Agent Orchestration, Vector*

### Mistral 7B Fine-tuned Chatbot · Personal Project

- Fine-tuned Mistral 7B LLM using LoRA (Low-Rank Adaptation) adapters for parameter-efficient fine-tuning, implementing quantization to optimize memory usage without sacrificing model
- Demonstrated hands-on experience with open-source LLM fine-tuning workflows, model quantization, and custom development outside of enterprise

*Tech Stack: Python, Mistral 7B, LoRA, HuggingFace Transformers, bitsandbytes (4-bit quantization),*

### Logistics Document AI · Personal Project

- Built a production-style FastAPI application enabling users to upload logistics PDF documents and ask natural language via a full RAG pipeline — retrieval, LLM answer generation, confidence scoring, and
- Implemented structured shipment data extraction (shipper, consignee, carrier, dates, rates) from unstructured PDFs, OCR-style parsing with LLM-based field
- Designed confidence scoring and guardrail logic to validate LLM responses before returning to the client, ensuring document-sensitive
- Demonstrates end-to-end ownership from PDF ingestion to frontend rendering via Jinja2
- GitHub: [github.com/Monishka1410/logistics-document-](https://github.com/Monishka1410/logistics-document-)

*Tech Stack: Python, FastAPI, RAG, Vector Embeddings, LangChain, LLM, PyPDF, Jinja2,*

## EDUCATION

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### Bachelor of Engineering – Electronics

*Shri Ramdeobaba College of Engineering, Nagpur, India Aug 2019 – May*